

Advanced Algorithms COMP4121, Term 3 of 2021

FINAL EXAM QUESTIONS

Exam closes at 5:30. If you are a registered ELS student, your exam closes at 6:30. Submit your typed or **CLEARLY WRITTEN** scanned answers by email to ignjat@cse.unsw.edu.au. To email, use **ONLY** your CSE email account. If you are scanning make sure your file is readable and less than 5MB in size, otherwise it might not go through. If you have any questions email me and I will try to answer them as promptly as I can. Problems are worth as indicated. You can use freely my slides. There are 5 questions on 3 pages. You must show your work and justify your answers. Good luck!

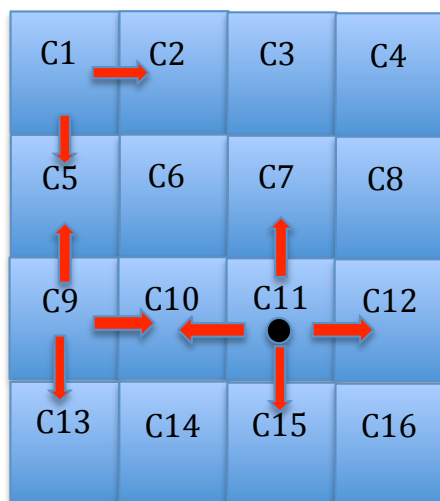
1. Assume you are using a biased coin to construct a Skip List. The probability of getting a **head** is $1/3$ and probability of getting a **tail** is $2/3$. When deciding the number of levels at which you will have a link for an element to be inserted you keep tossing your biased coin until you get a tail and the number of levels with links is equal to the number of **heads** you obtain before getting a tail.
 - (a) Estimate the probability that an element has links on levels $0 - i$ (and possibly also on higher levels). **(3 points)**
 - (b) Assume such a Skip List has n elements; estimate the expected number of elements with links on level i . **(3 points)**
 - (c) Estimate the probability to have at least one element on level $k \log_3 n$ where k is an arbitrary integer $k \geq 1$. **(3 points)**
 - (d) Estimate the expected value E of k such that k is the least integer so that the number of levels does not exceed $k \log_3 n$. **(3 points)**
 - (e) Estimate the probability that an element with links up to level i also has a link at level $i + 1$. **(3 points)**

- (f) Estimate the expected number of elements between any two consecutive elements with a link on level $i + 1$ which have links on all levels from 0 to i but not on level $i + 1$. **(3 points)**
- (g) Estimate the search time in such a SkipList. **(3 points)**
- (h) So what do you think, is searching in such a SkipList built with such a biased coin asymptotically faster or slower than searching in the usual SkipList built with an unbiased coin? **(3 points)**

(Each item is worth 3 points so in total the whole problem is worth 24 points.)

2. Assume that you have a device which produces random lists of 10,000 uniformly distributed random integers in the range $0 - 10^{10}$. You need to sort 10,000 of such lists and have an option to use either the QuickSort or the Randomised Quick Sort. Which one would do the job faster? Explain why. **(10 points)**
3. You have a randomised algorithm which on an input of length n succeeds in certain task with a probability of $1/n$. You can run it $k(n) \times n$ many times, where $k(n)$ is a function you have to choose. What should you choose for $k(n)$ to succeed at least once with probability of at least $1 - 1/n$? Justify your answer by showing your calculations. **(16 points)**
4. Consider a 4X4 board consisting of 16 squares with a spider walking on it; see the figure on the next page. Every second the spider moves from the square where it is at the moment to any of the available adjacent squares vertically up or down or horizontally left or right (but not diagonally and not off the board) with equal probability in all of the available directions. Consider the Markov process whose states are the 16 squares where the spider can be at each second.
 - (a) Consider the transition probabilities which corresponds to such a Markov chain with states enumerated as shown on the figure so that the transition probabilities satisfy the stated requirements. Specify the transition probabilities:
 - i. From C1 to all other states. **(2 points)**
 - ii. From C2 to all other states. **(2 points)**
 - iii. From C6 to all other states. **(2 points)**
 - (b) Is such a Markov Chain irreducible, i.e., is the underlying graph strongly connected? **(8 points)**

- (c) Is such a Markov Chain periodic? (this is a tiny bit tricky so think carefully if the length of every path starting and ending at the same square must be divisible by a fixed number) **(16 points)**



5. Assume you have a ball of unit radius in a space of a very high dimension d , say $d \geq 1000$. You draw a point from such a unit ball uniformly at random. Show that the probability that such a point will be at a distance at most $1/d$ from the surface of such a ball is larger than $1 - 1/e$ where e is the Euler constant $2.71\dots$ **(20 points)**